DO NOT LEAN
OVER SIDE OF
RAILINGS

Analysing the factors affecting Railway Accidents using Big Data

Background and Context

- Railway systems are crucial for global infrastructure, facilitating economic growth and efficient transportation.
- Despite their advantages, railway accidents pose significant risks due to factors like human error and mechanical failures.
- Modern technologies like Big Data and AI offer promising solutions to enhance railway safety.

Background and Problem Statement

- Railway accidents result from human errors, mechanical failures, environmental conditions, and infrastructural issues.
- Traditional accident analysis methods are retrospective and lack real-time predictive capabilities.
- Big Data analytics and AI offer a new approach by analyzing vast data to predict accidents and enhance safety proactively.

Research Questions

1. What are the primary factors contributing to railway accidents?
2. How can Big Data analytics be utilized effectively?
3. What patterns or trends exist in historical accident data?
4. How can AI predict railway accidents?
5. What are the components of an AI-based alert system?
6. How effective is the AI-based system in reducing accidents?

Research Aim and Objectives

Research Aim:

Enhance railway safety using an AI-based system to predict and prevent accidents through Big Data analytics.

Objectives:

1. Identify and classify factors contributing to railway accidents.
2. Collect and preprocess historical and real-time railway accident data.
3. Apply Big Data analytics to identify significant patterns in accident data.
4. Develop machine learning models to predict potential railway accidents.
5. Design an AI-based alert system for real-time accident prevention.
6. Evaluate the system's effectiveness through simulations and case studies.

Relevance and Importance of the Research

Relevance:

- Railway accidents, though infrequent, have severe consequences.
- Using Big Data and AI can proactively enhance railway safety.

Importance:

- Railway Operators
- Regulatory Bodies
- Policymakers
- Passengers
- Academic Contribution

Theoretical Framework Development

- Accident Causation Models
- Big Data Analytics Frameworks
- AI and Machine Learning Theories
- Risk Assessment and Management Theories
- Human Factors Engineering

Overview of Railway Accidents

Types of Railway Accidents:

- Collisions
- Derailments
- Level Crossing Incidents
- Natural Causes

Impact of Accidents:

- Loss of life, injuries, economic losses, service disruptions.
- Psychological impact on survivors and responders.
- Erosion of public confidence in railway safety.

Insights from Historical Data:

- Human error a significant factor (70% of accidents in Europe).
- Importance of training, safety technologies, improved working conditions.
- Need for robust track maintenance, modern signalling, resilient infrastructure.

Challenges and Opportunities of Using Big Data to Prevent Railway Accidents

Challenges:

- Data Quality and Integration
- Data Volume and Processing
- Privacy and Security
- Technical Infrastructure
- Skill Gaps

Opportunities

- Predictive Maintenance
- Real-Time Monitoring
- Enhanced Decision-Making
- Improved Passenger Experience
- Cost Savings

Role of Big Data in Railway Sector

- Predictive Maintenance
- Real-Time Monitoring
- Passenger Analytics
- Infrastructure Management
- Operational Efficiency

Case Studies and Examples

1. Deutsche Bahn
2. Indian Railways
3. Network Rail (UK)
4. Union Pacific Railroad (USA)
5. China Railway Corporation

Prediction Tasks in Railway Accidents

1. Predicting Equipment Failures
2. Forecasting Weather-Related Hazards
3. Identifying High-Risk Areas
4. Predicting Human Errors

Advantages and Disadvantages in Accident Prediction

Advantages:

1. Accuracy
2. Proactive Measures
3. Real-Time Analysis
4. Efficiency

Advantages:

1. Data Quality
2. Computational Resources
3. Complexity
4. Overfitting

Challenges in Implementing Prediction Algorithms

1. Data Integration
2. Model Interpretability
3. Scalability
4. Regulatory Compliance
5. Technical Infrastructure

Key Findings and Recommendations for Prediction Algorithm Applications in the Railway Sector

- Data Quality and Integration
- Algorithm Selection
- Real-Time Capabilities
- Skill Development
- Interdisciplinary Collaboration

Gap Analysis and Novelty of the Research

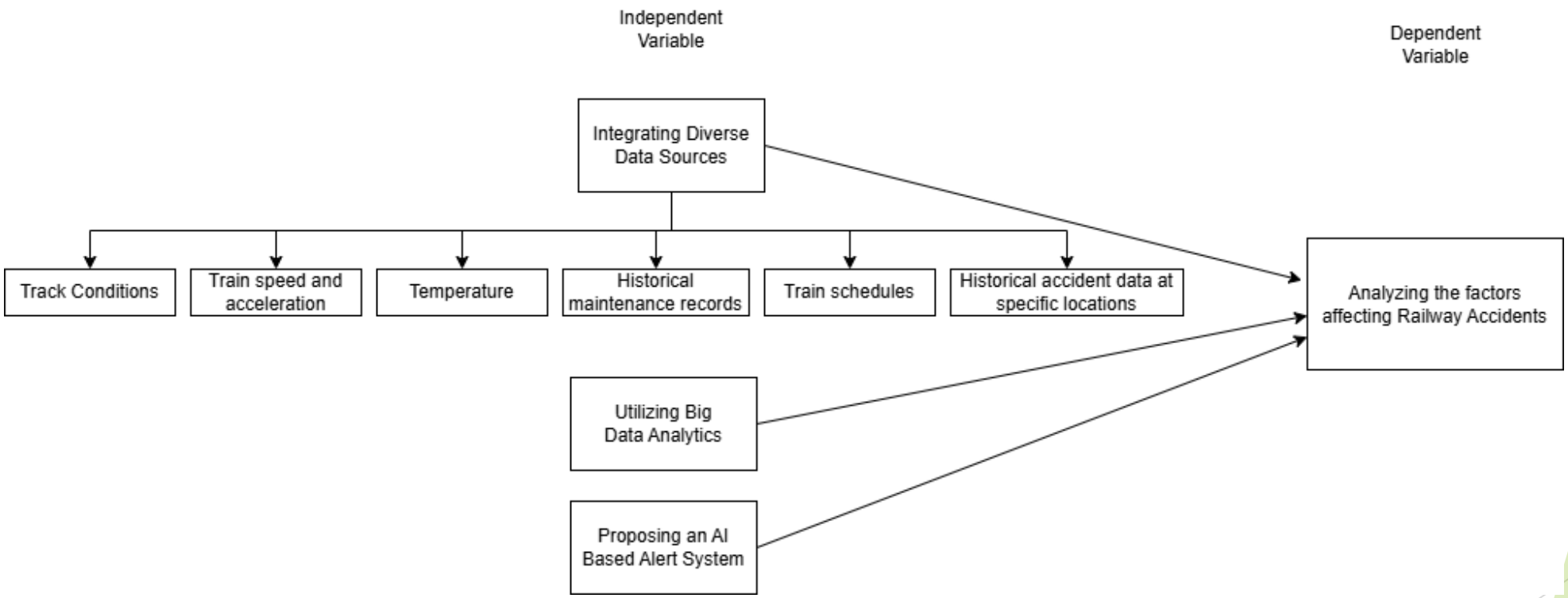
Current Gaps in Research:

1. Integration of Diverse Data Sources
2. Real-Time Predictive Analytics
3. AI-Based Alert Systems

Novelty of the Research:

1. Integrated Approach
2. Real-Time Predictive Analytics Framework
3. AI-Based Alert System

Conceptual Diagram



Hypothesis Development

1. Hypothesis 1: Integrating Diverse Data Sources
2. Hypothesis 2: Real-Time Predictive Analytics
3. Hypothesis 3: AI-Based Alert System

Research Design

Descriptive Research

- ✓ Establish baseline conditions, identify effective practices, and gaps for enhancement.

Exploratory Research

- ✓ Identify relationships, patterns, and correlations in Big Data for proactive safety measures.

Integration of Methods

- ❖ Qualitative Methods:
  Analyse textual data (reports, interviews, literature) to understand contextual challenges.

- ❖ Quantitative Methods:
  Use statistical analysis and machine learning on sensor, maintenance, and accident data.

Data Collection

- Introduction to Data Collection
- Interview Insights
- Primary Dataset
- Supplementary Data Sources

Data Analysis

Data Preprocessing

- Handling Missing Values
- Normalizing Data
- Removing Outliers
- Data Transformation

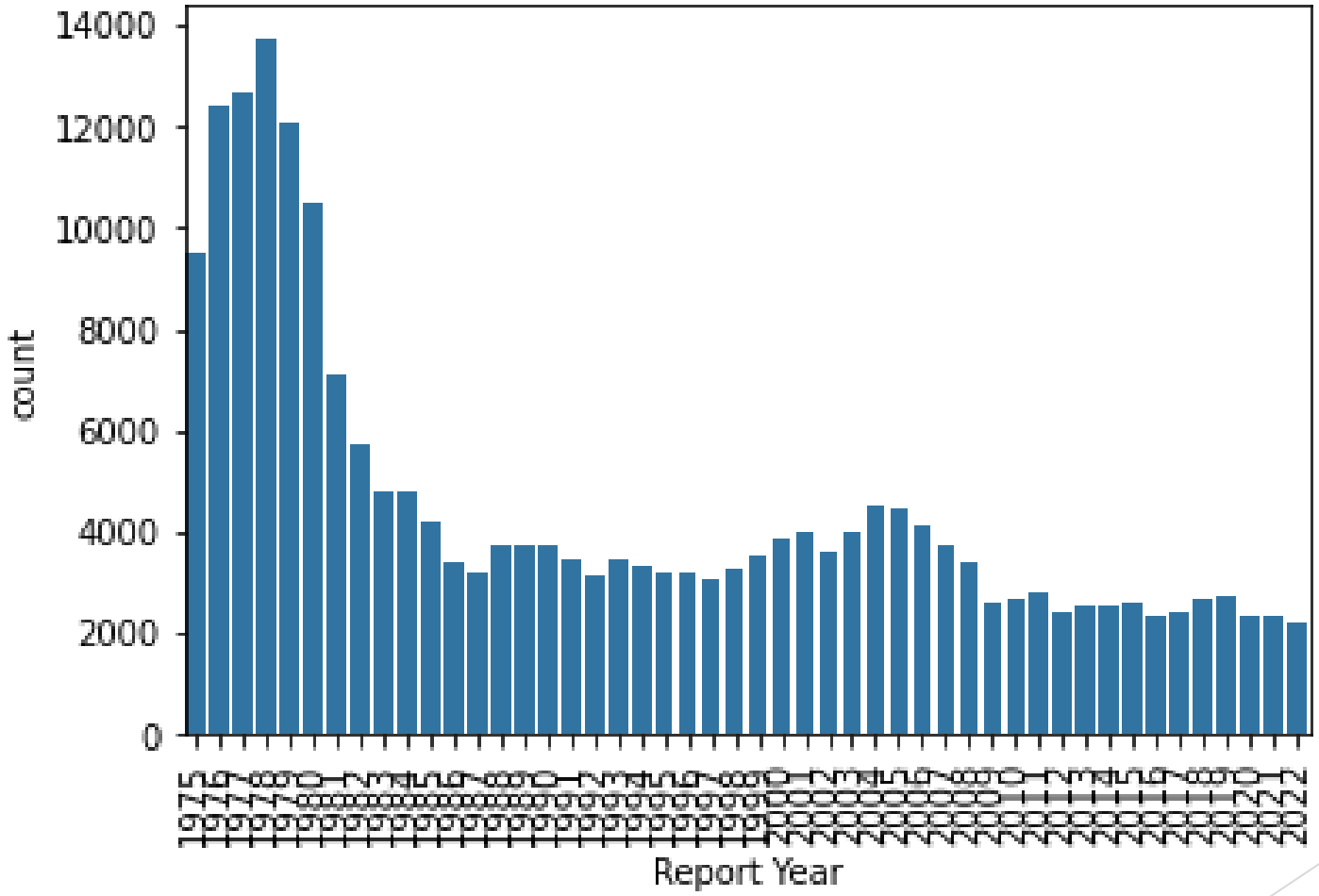
Data Analysis

Exploratory Data Analysis (EDA)

- Descriptive Statistics
- Data Visualization
- Correlation Analysis
- Distribution Analysis
- Identifying Patterns

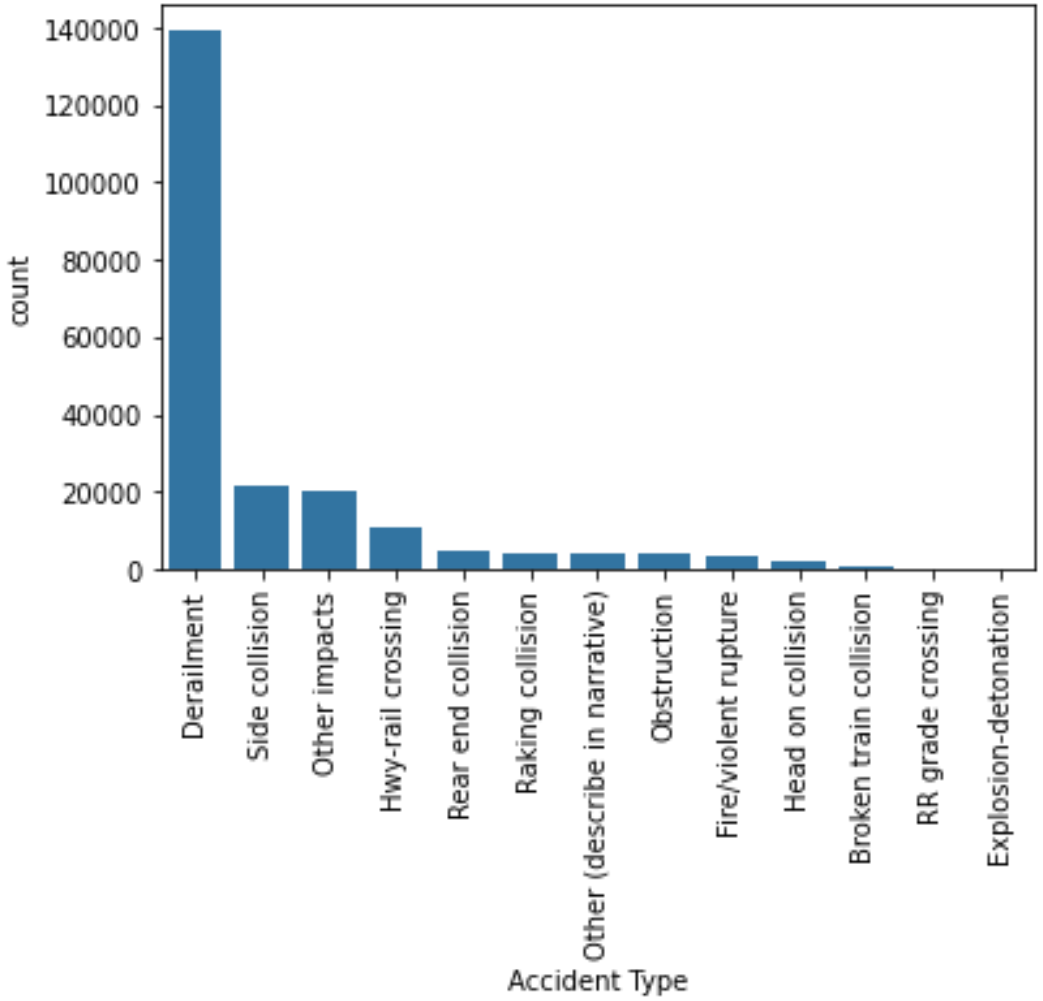
Data Analysis

Data Coverage Across Years



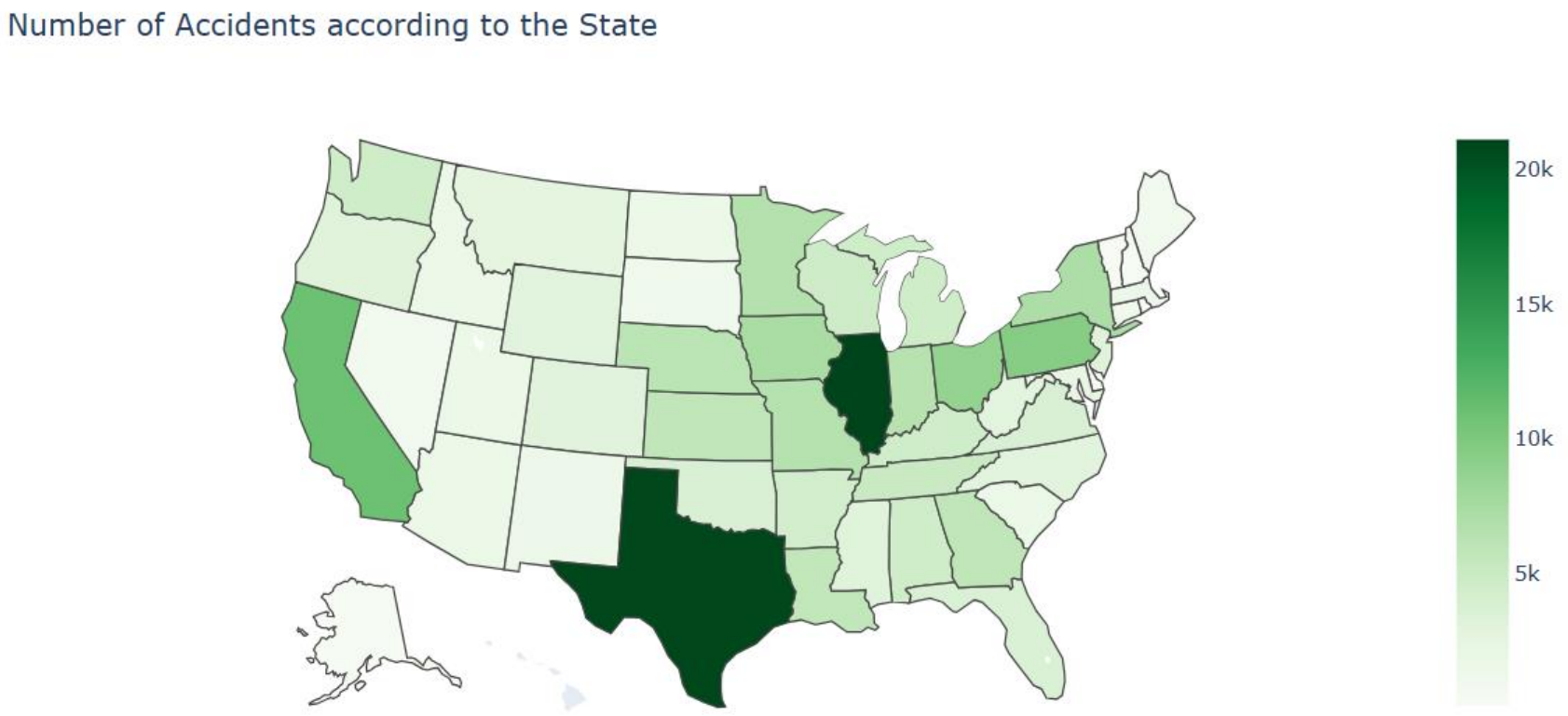
Data Analysis

Most Common Accident Types



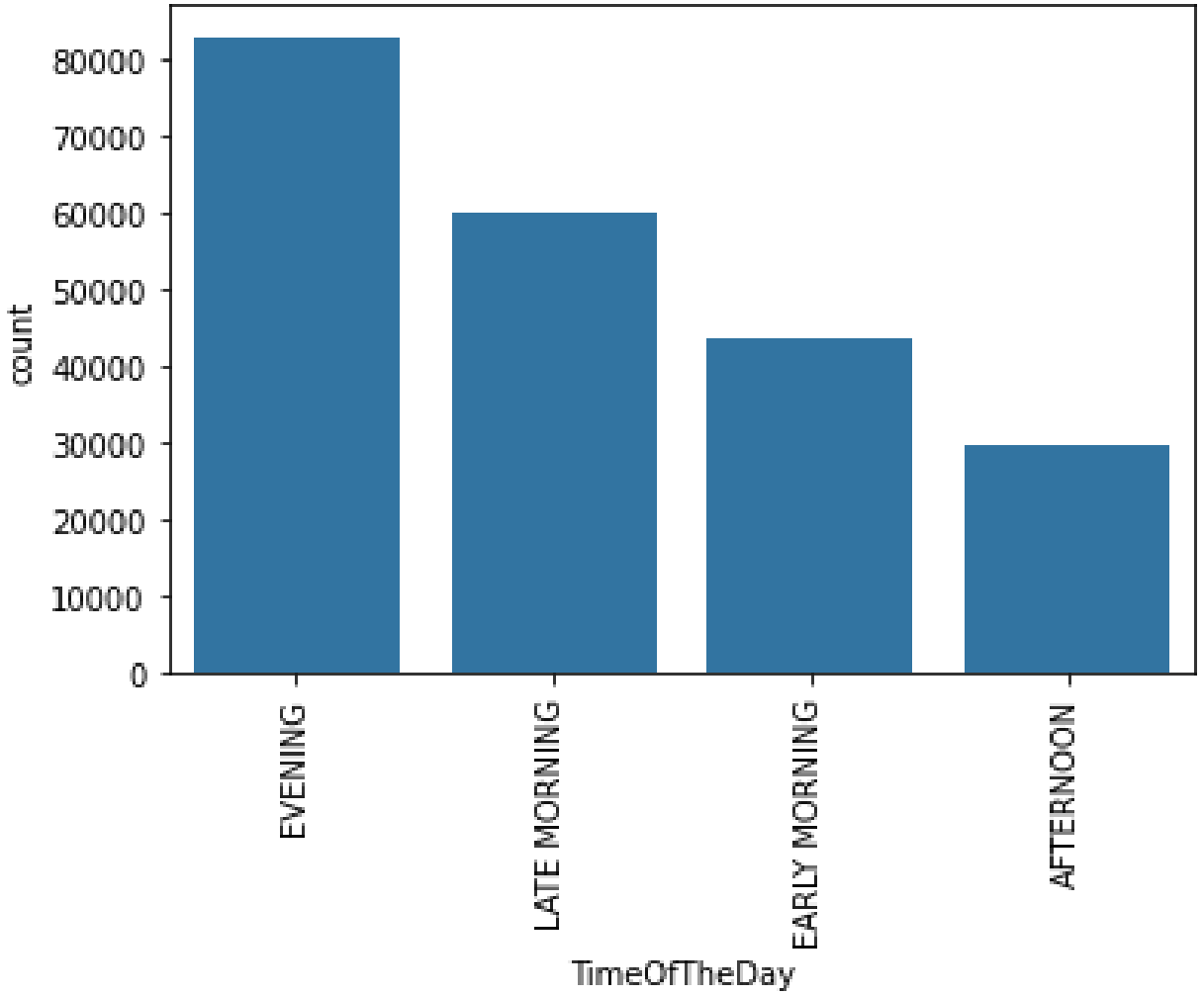
Data Analysis

Number of Accidents according to the State(Map)



Data Analysis

Times of the day where most of the accident occur



Data Analysis

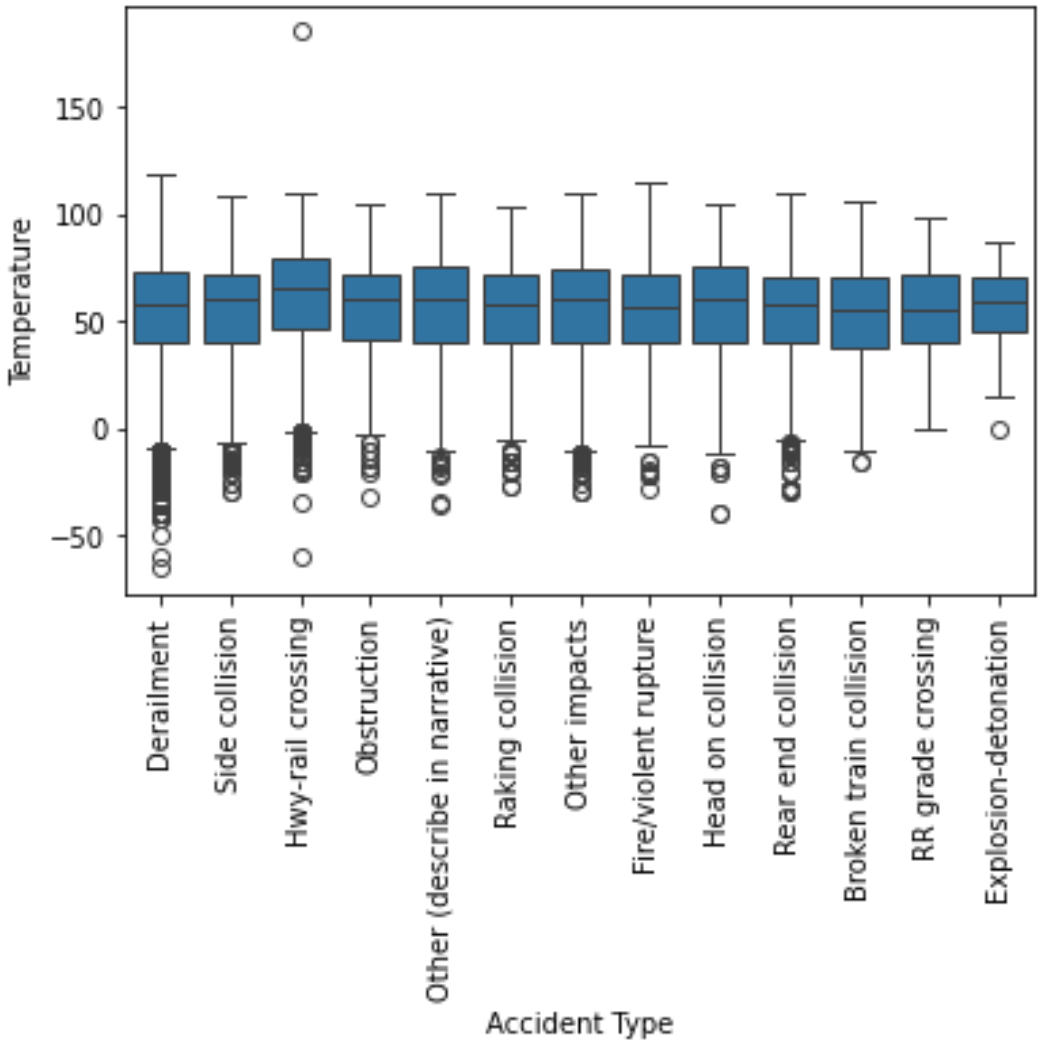
Correlation between variables

Out[30]:

	First Variable	Second Variable	Coefficient Value
0	Accident Year	Maintenance Accident Year	1.000000
1	Accident Month	Maintenance Accident Month	1.000000
2	Maintenance Accident Year	Accident Year	1.000000
3	Maintenance Accident Month	Accident Month	1.000000
4	Train Speed	Maximum Speed	0.881809
5	Maximum Speed	Train Speed	0.881809
6	Persons Killed For Reporting Railroad	Total Killed Form 54	0.886141
7	Persons Injured For Reporting Railroad	Total Persons Injured	0.903628
8	Total Persons Injured	Persons Injured For Reporting Railroad	0.903628
9	Total Killed Form 54	Persons Killed For Reporting Railroad	0.886141

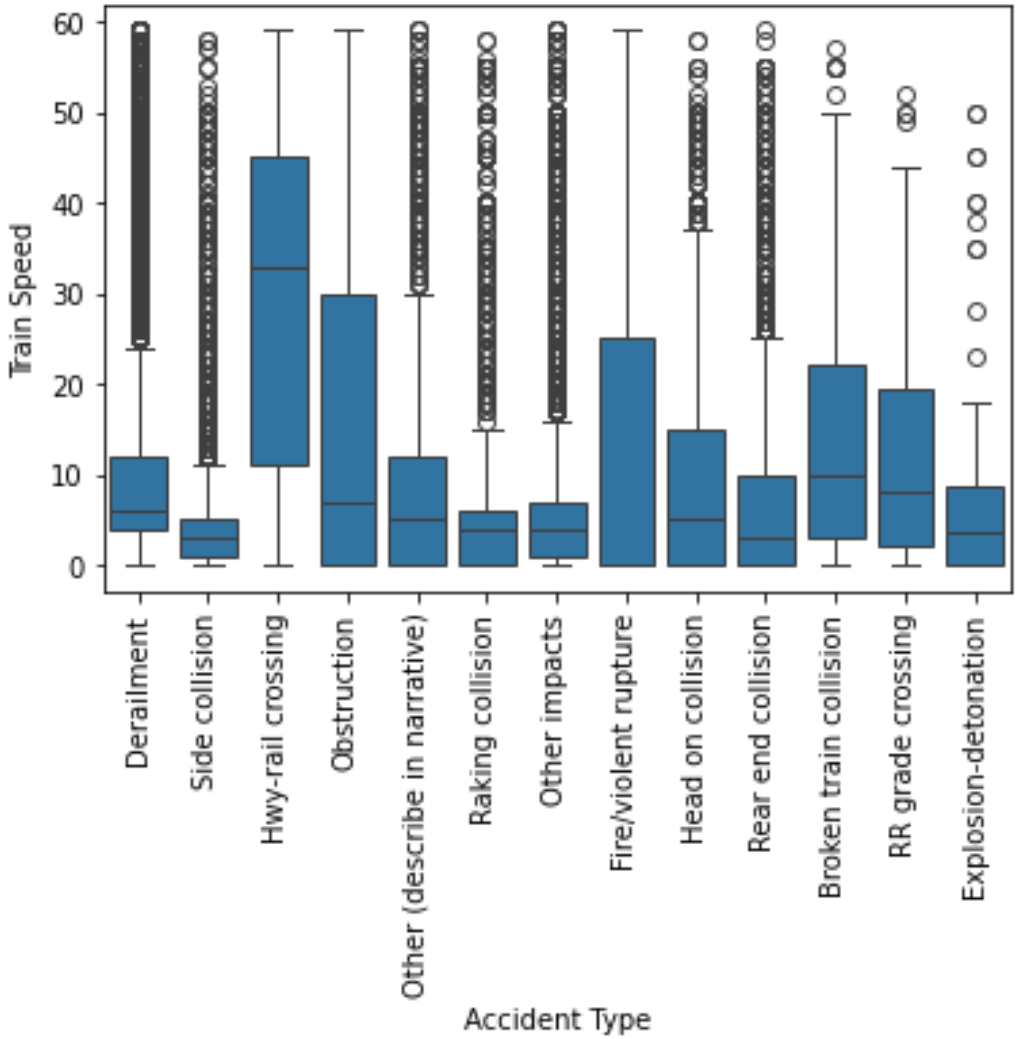
Data Analysis

Relationship between Temperatures and Accident Types

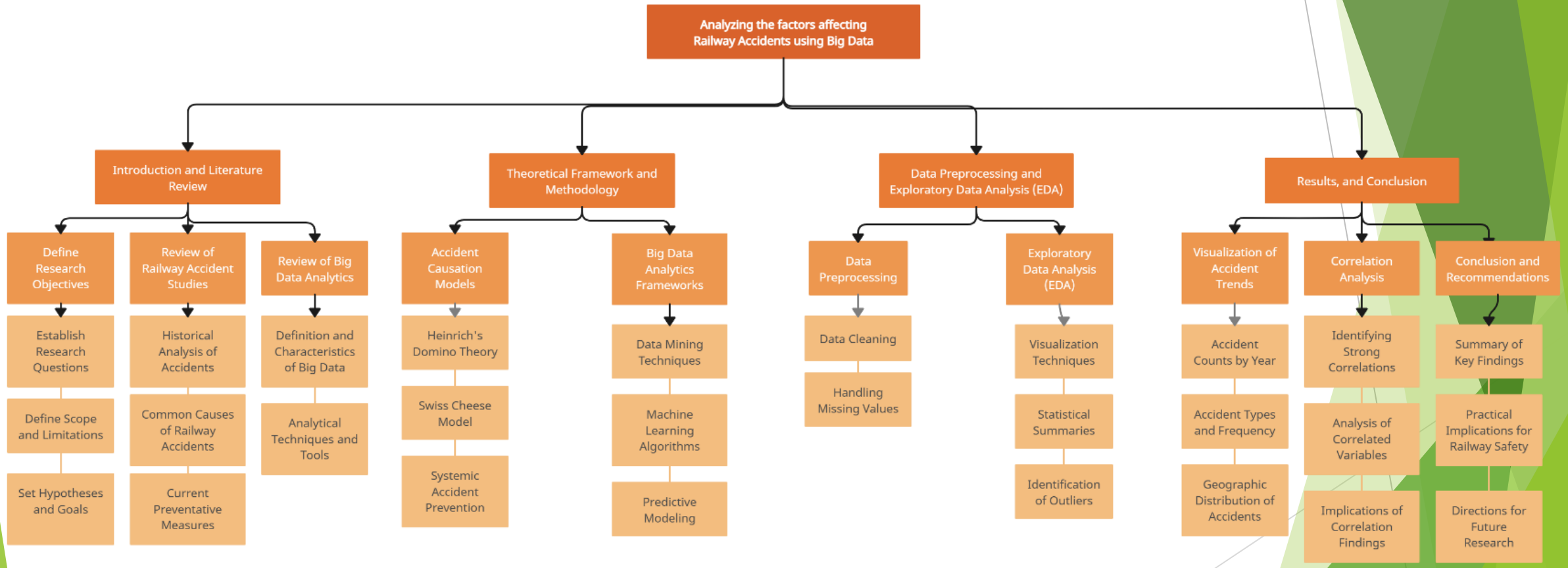


Data Analysis

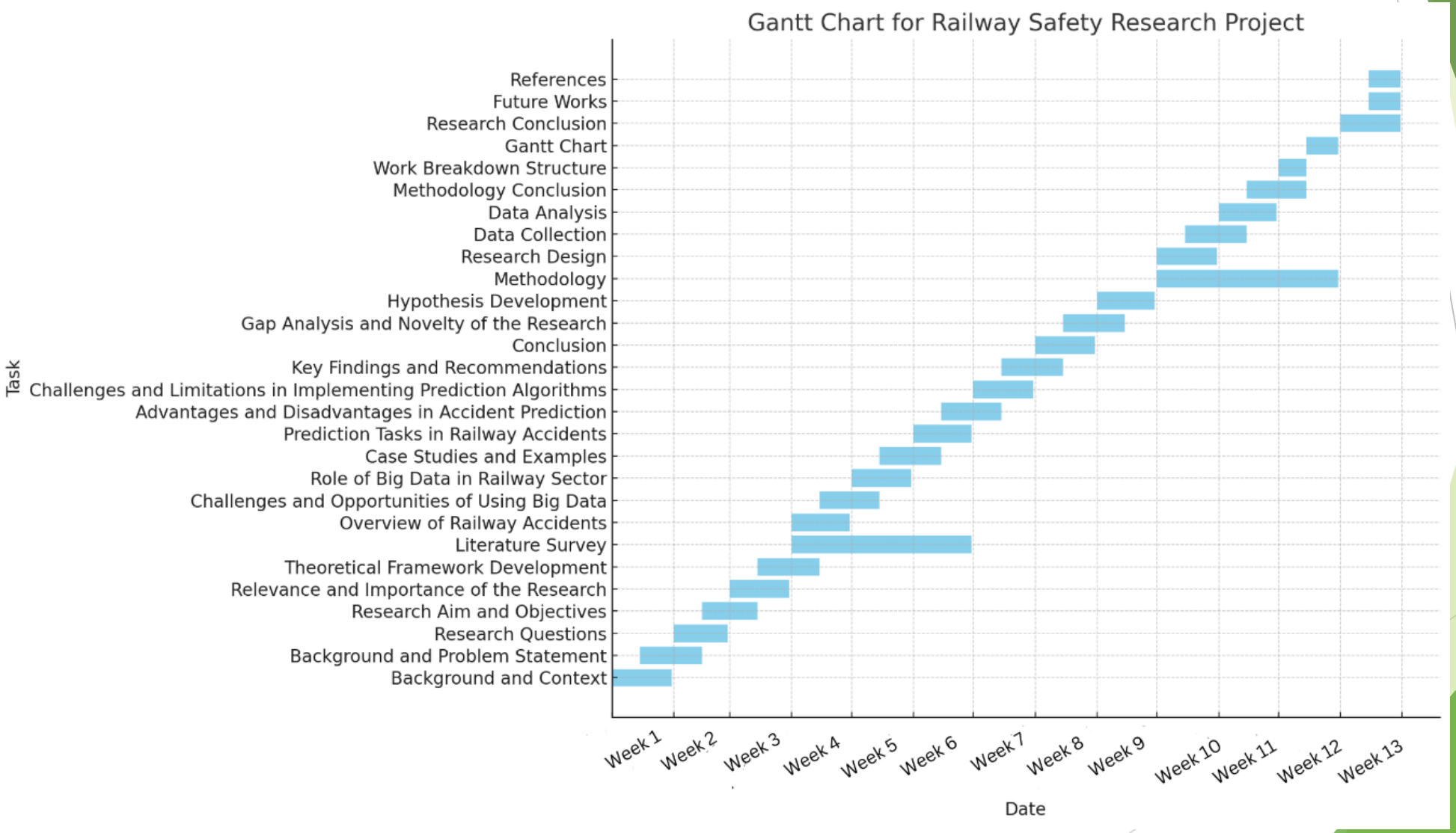
Relationship between Train Speeds and Types



Work Breakdown Structure



Gant Chart


Gantt Chart for Railway Safety Research Project
References
Future Works
Research Conclusion
Gantt Chart
Work Breakdown Structure
Methodology Conclusion
Data Analysis
Data Collection
Research Design
Methodology
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Background and Context
Task
Week 1
Week 2
Week 3
Week 4
Week 5
Week 6
Week 7
Week 8
Week 9
Week 10
Week 11
Week 12
Week 13
Date

Future Works

1. System Architecture and Prediction Models

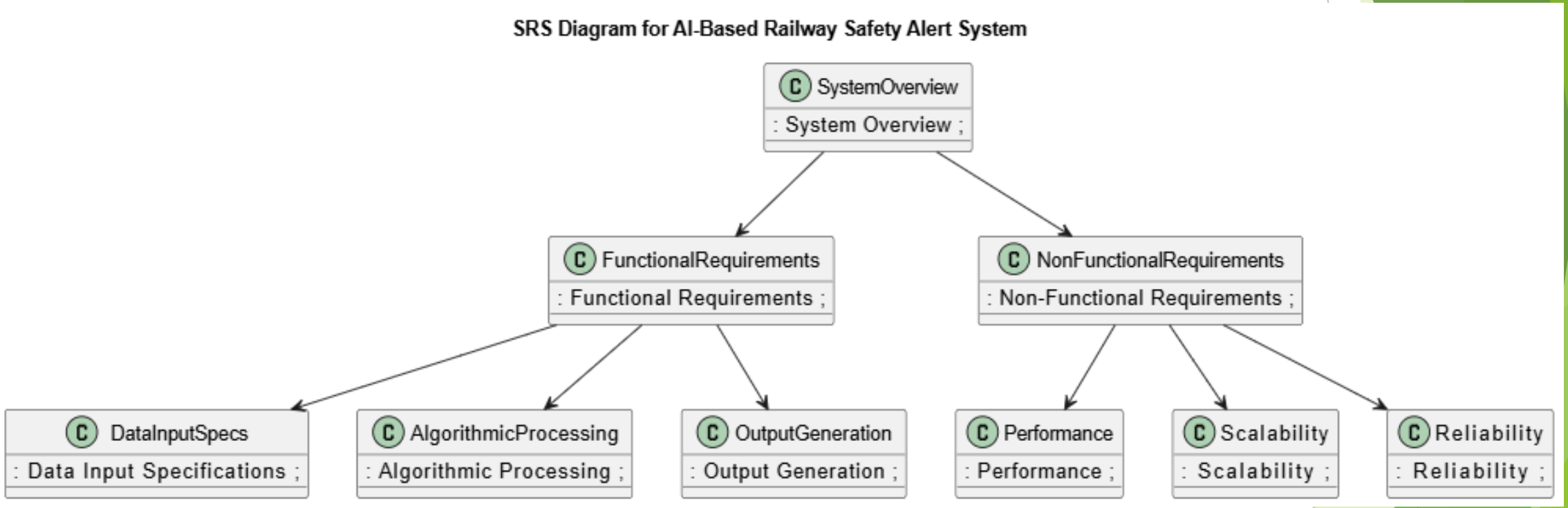
- ✓ AI-Based Alerting System
- ✓ Real-Time Anomaly Detection

2. Features of the Proposed System

- ✓ Real-Time Monitoring
- ✓ Predictive Analytics
- ✓ Anomaly Detection
- ✓ User Alerts
- ✓ Data Visualization

Use case Diagram

SRS Diagram


SRS Diagram for AI-Based Railway Safety Alert System
SystemOverview
: System Overview ;
FunctionalRequirements
: Functional Requirements ;
NonFunctionalRequirements
: Non-Functional Requirements ;
DataInputSpecs
: Data Input Specifications ;
AlgorithmicProcessing
: Algorithmic Processing ;
OutputGeneration
: Output Generation ;
Performance
: Performance ;
Scalability
: Scalability ;
Reliability
: Reliability ;

Conclusion

Thank You

- Special Thanks for my supervisor Mr. Deshan.